1. Connected Particles

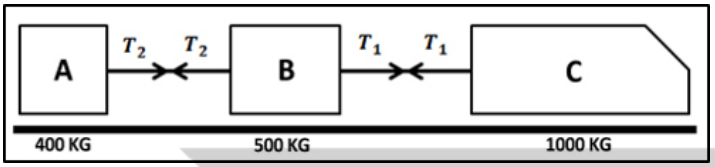
1.1. Introduction

Newton's 3rd Law of Motion:

For every action, there is an equal and opposite reaction

Question:

A train pulls two carriages:



The forward force of the engine is $F = 2500N$. Find the acceleration and tension in each coupling. The resistance to motion of A, B, and C are 90, 150, and 200N, respectively.

Solution:

To find acceleration, regard the system as a single object. The internal T s cancel out and give:

$$2500 - (200 + 150 + 90) = 1900a$$

$$\therefore a = 1.08ms^{-2}$$

To find T_1, look at C

$$F - T_1 - 200 = 1000a$$

$$2500 - T_1 - 200 = 1000 \times 1.08$$

$$T_1 = 1220N$$

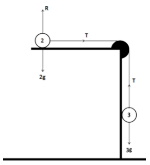
To find T_2, look at A

$$T_2 - 90 = 400a$$

$$T_2 - 90 = 400 \times 1.08$$

$$T_1 = 522N$$

1.2. Pulleys



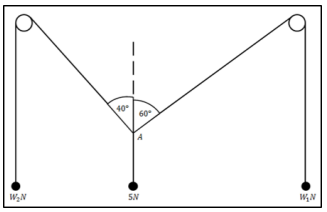
- Equation 1:

$$\therefore T = 2a$$

- Equation 2:

$$3g - T = 3a$$

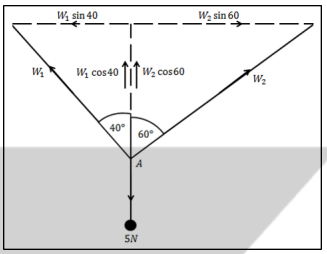
{W05-P04} Question 3:



The strings are in equilibrium. The pegs are smooth. All the weights are vertical. Find W_1 and W_2

Solution

Diagram showing how to resolve forces:



Resolving forces at A vertically:

$$W_1 \cos 40 + W_2 \cos 60 = 5$$

Resolving forces at A horizontally:

$$W_1 \sin 40 = W_2 \sin 60$$

Substitute second equation into first:

$$\left(\frac{W_2 \sin 60}{\sin 40}\right) \cos 40 + W_2 \cos 60 = 5$$

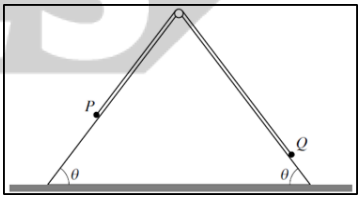
Solve to find W_2 :

$$W_2 = 3.26N$$

Put this value back into the first equation to find W_1

$$W_1 = 4.40N$$

{S12-P41} Question 6:



P has a mass of $0.6kg$ and Q has a mass of $0.4kg$. The pulley and surface of both sides are smooth. The base of triangle is horizontal. It is given that $\sin\theta = 0.8$. Initially particles are held at rest on slopes with string taut. Particles are released and move along the slope

1. Find tension in string. Find acceleration of particles while both are moving.

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2. Speed of P when it reaches the ground is $2ms^{-1}$. When P reaches the ground, it stops moving. Q continues moving up slope but does not reach the pulley. Given this, find the time when Q reaches its maximum height above ground since the instant it was released

Solution

Part (i)

Effect of weight caused by P in direction of slope:

Effect of weight $= mg\sin\theta$ where $\sin\theta = 0.8$

Effect of weight $= 4.8N$

Effect of weight caused by Q in direction of slope:

Effect of weight $= 0.4 \times 10 \times 0.8 = 3.2N$

Body P has greater mass than body Q so when released P moves down Q moves up on their slopes $\therefore$

$$4.8 - T = 0.6a$$

$$T - 3.2 = 0.4a$$

Solve simultaneous equations:

$\frac{4.8-T}{0.6} = \frac{T-3.2}{0.4}$ $\quad T = 3.84N$

Substitute back into initial equations to find a:

$4.8 - 3.84 = 0.6a \quad a = 1.6ms^{-2}$

Part (ii)

Use kinematics equations to find the time which it takes P to reach the ground:

$a = \frac{v-u}{t}$ and $t = \frac{2-0}{1.6}$

$$t_1 = 1.25s$$

When P reaches the ground, only force acting on Q is its own weight in the direction of slope $= 3.2N$

$F = ma \quad -3.2 = 0.4a$

$$a = -8ms^{-2}$$

Now calculate the time taken for Q to reach max height

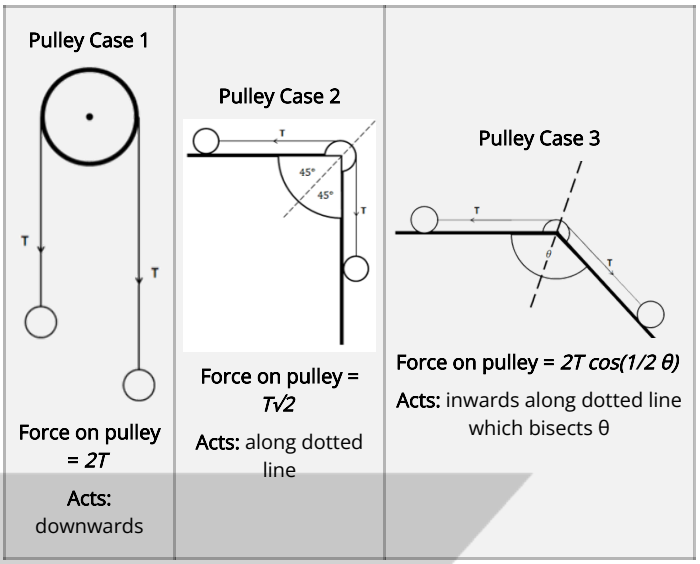
This occurs when its final velocity is 0.

$$-8 = \frac{0-2}{t} \quad t_2 = 0.25s$$

Now do simple addition to find total time:

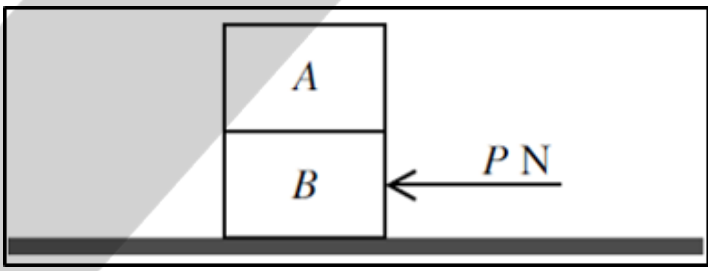
Total Time $= 1.25 + 0.25 = 1.5s$

1.3. Force Exerted by String on Pulley



1.4. Two Particles

{S10-P43} Question 7:



A and B are rectangular boxes of identical sizes and are at rest on rough horizontal plane. A mass $= 200kg$ and B mass $= 250kg$. If $P \leq 3150$ boxes remain at rest. If $P > 3150$ boxes move

1. Find coefficient of friction between B and floor
2. Coefficient of friction between boxes is 0.2. Given that $P > 3150$ and no sliding occurs between boxes. Show that the acceleration of boxes is not greater than $2ms^{-2}$
3. Find the maximum possible value of P in the above scenario

Solution

Part (i)

$$F = \mu N$$

$F =$ to max P that does not move the boxes

$N =$ to contact force of **both** boxes acting on floor

$$\therefore 3150 = \mu \times (2000 + 2500)$$

$$\mu = 0.7$$

Part (ii)

Find frictional force between A and B:

$F = 0.2 \times 2000 \quad F = 400N$

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Use Newton's Second Law of Motion to find max acceleration for which boxes do not slide (below F)

$400 = 200a \quad a = 2ms^{-2}$

Part (iii)

P has to cause an acceleration of $2ms^{-2}$ on B which will pass on to A as they are connected bodies

Simply implement Newton's Second Law of Motion

$$\therefore P = (200 + 250)(2) + 3150$$

The 3150 comes from the force required to overcome the friction

$P = 900 + 3150 \quad = 4050N$

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